

[Academic Presentation Title]

[Presentation Subtitle]

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[Conference Name, Month Year]

Progressive vs. Non-progressive Visualization

Non-progressive visualization

- Renders in one pass.
- Each change in the data, whether caused by interaction or new data, triggers a full rendering.

Progressive visualization

- Rendering uses a series of passes that improve the result until completion.
- Each pass is limited to a requested latency and renders a meaningful partial result.
- Each partial result has a meaningful error or quality measure and an estimate of completion, such as a progress bar.
- Users can interact at any time; the visualization adapts as much as possible and can implement steering when needed.

Definition adapted from Ulmer et al., “A Survey on Progressive Visualization” (2024).

Progressive Computation

$$F_p(P, D_1, q, \emptyset) \xrightarrow{t_1, m_1} R_1, \quad \dots,$$

$$F_p(P, D_z, q, R_{z-1}) \xrightarrow{t_z, m_z} R_z$$

$t_i \leq q$, $R_i \rightarrow R$ or the computation reaches a termination criterion.

F_p : progressive function

P : function parameters

R_{i-1} : previous result; R_i : partial result

$\emptyset$: no previous result at the first pass

q : requested interval between results

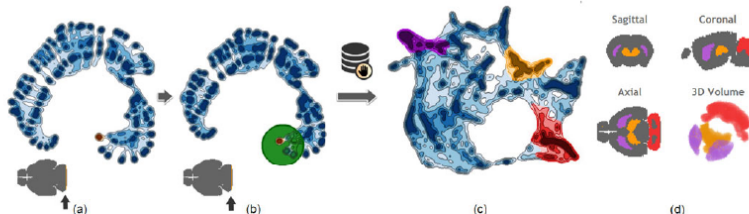
D_i : data processed at pass i

t_i, m_i : time and memory at pass i

R : final result

Equations: Fekete & Primet (2016) [1], as cited by Ulmer et al. (2024) [3].

Image: A-tSNE mouse-brain case study from Pezzotti et al. (2017) [2].



Requirements for Progressive Visualization

- Computation proceeds in bounded passes, each returning a meaningful partial result.
- Partial results should communicate quality or error and give an estimate of completion.
- Users can interact before completion; steering can direct further computation toward relevant regions.
- Data chunking addresses large inputs; process chunking exposes progress in slow algorithms.

Sources: Ulmer et al. (2024); Pezzotti et al. (2017).

A Taxonomy of Progressive Visualizations

- **Visualization type:** temporal, geo-spatial, hierarchical, network, multidimensional, or field.
- **Processing:** data chunking, process chunking, or custom chunking.
- **Data domain:** known end or unknown end.
- **Visual update:** extend the current view or overwrite it with a new state.

Source: Ulmer et al. (2024).

Progressive Dimensionality Reduction

- t-SNE builds an embedding through iterative optimization, but it must first compute high-dimensional similarities.
- Barnes-Hut t-SNE reduces optimization cost, yet initialization can still delay the first interpretable embedding.
- Q-SNE approximated neighbors for sparse document data; A-tSNE extends approximate similarities to dense high-dimensional data.

Source: Pezzotti et al. (2017), Related Work.

Field Trends and Open Challenges

- The survey classifies 48 papers from 1999 to 2023; 33 of 48 use multidimensional visualization.
- Steering appears in 35 studies, uncertainty in 20, and real-time processing in 10.
- Open challenges include visual stability as estimates change, uncertainty communication, and evaluation beyond runtime.

Source: Ulmer et al. (2024), study set and discussion.

Synthesis and Paper Positioning

- A-tSNE addresses an algorithm-level bottleneck: delayed initialization in high-dimensional t-SNE.
- The 2024 survey offers a field-level taxonomy and reviews evaluation methods and research gaps.
- **Synthesis:** progressive methods must make partial results timely, interpretable, steerable, and stable as they improve.

Sources: Pezzotti et al. (2017); Ulmer et al. (2024).

Paper 1: Overview

- *[Title, authors, and publication details]*
- *[Research subject and context]*
- *[Core question addressed in this paper]*

Paper 1: Motivation

- *[Why the research question matters]*
- *[Limitations of existing explanations]*
- *[How the paper approaches the question]*

Paper 1: Research Questions and Hypotheses

- *[Research question]*
- *[Hypothesis or proposition 1]*
- *[Hypothesis or proposition 2, if applicable]*

Paper 1: Conceptual Framework

- *[Key concepts or variables]*
- *[Theoretical relationships and proposed mechanisms]*
- *[Insert the paper's central framework figure here]*

Paper 1: Data and Research Subjects

- *[Data or material sources and time period]*
- *[Sample composition or case selection]*
- *[Data processing and ethics, if applicable]*

Paper 1: Research Methods

- *[Research design and analytical method]*
- *[Model, experiment, or analysis workflow]*
- *[Key identification assumptions or derivation conditions]*

Paper 1: Results Overview

- *[Main result 1]*
- *[Main result 2]*
- *[How the results address the hypotheses or propositions]*

Paper 1: Main Evidence

- *[Insert the key figure, table, or theorem]*
- *[Key estimate, proof, or observation]*
- *[Conclusion supported by this evidence]*

Paper 1: Mechanisms and Additional Analyses

- *[Mechanism analysis or theoretical explanation]*
- *[Robustness, sensitivity, or alternative explanation checks]*
- *[Boundary conditions for the results]*

Paper 1: Summary

- *[Main conclusion]*
- *[Contribution and limitation]*
- *[Connection to Paper 2]*

Paper 2: Overview

- *[Title, authors, and publication details]*
- *[Research subject and context]*
- *[Core question addressed in this paper]*

Paper 2: Research Question and Contribution

- *[Research question]*
- *[What this paper adds to related work]*
- *[Relationship to or distinction from Paper 1]*

Paper 2: Framework and Hypotheses

- *[Theoretical basis or conceptual framework]*
- *[Hypotheses or propositions]*
- *[Proposed mechanism or reasoning]*

Paper 2: Data and Research Design

- *[Data or material sources and sample]*
- *[Research design and key variables or concepts]*
- *[Why the design fits the research question]*

Paper 2: Methods and Analysis Strategy

- *[Model, experiment, comparison, or proof method]*
- *[Identification strategy or analytical steps]*
- *[Key assumptions and how they are assessed]*

Paper 2: Main Results

- *[Main result 1 and supporting evidence]*
- *[Main result 2 and supporting evidence]*
- *[How the results answer the research question]*

Paper 2: Mechanisms and Robustness

- *[Mechanisms, heterogeneity, or boundary conditions]*
- *[Robustness or sensitivity analyses]*
- *[Alternative explanations and additional evidence]*

Paper 2: Summary and Overall Conclusions

- *[Main conclusion of Paper 2]*
- *[Shared findings and differences across the two papers]*
- *[Overall answer to the research topic]*

- [1] J.-D. Fekete and R. Primet, “Progressive Analytics: A Computation Paradigm for Exploratory Data Analysis,” CoRR, abs/1607.05162, 2016.
- [2] N. Pezzotti et al., Approximated and User Steerable t-SNE for Progressive Visual Analytics, *IEEE Transactions on Visualization and Computer Graphics*, 23(7), 1739–1752, 2017. doi:10.1109/TVCG.2016.2570755.
- [3] A. Ulmer et al., A Survey on Progressive Visualization, *IEEE Transactions on Visualization and Computer Graphics*, 30(9), 6447–6467, 2024. doi:10.1109/TVCG.2023.3346641.

[Thank You]

[Questions and Discussion]

[Presenter email, optional]